

A MINI PROJECT REPORT

on

SHEDURA

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

by

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Methodist College of Engineering and Technology,

King Koti, Abids, Hyderabad-500001.

2025-2026



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DECLARATION BY THE CANDIDATES

We, MOHD AYAAN (160724748089) and SYED AZEEM SADIQ (160724748102) students of Methodist College of Engineering and Technology, pursuing Bachelor's degree in Artificial Intelligence and Machine Learning, hereby declare that this Mini Project report entitled "SHEDURA", carried out under the guidance of Ms.J.Harika submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning. This is a record work carried out by us and the results embodied in this report have not been reproduced/copied from any source.

MOHD AYAAN (160724748089)

SYED AZEEM SADIQ (160724748102)



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CERTIFICATE BY THE SUPERVISOR

This is to certify that this Mini Project work entitled “SHEDURA” by Mohd Ayaan (160724748089) and Syed Azeem Sadiq (160724748102) submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning, during the academic year 2025-2026, is a bonafide record of work carried out by them.

Date:

Ms .J. Harika
Assistant Professor, Dept. of CSE(AI&ML).



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CERTIFICATE BY HEAD OF THE DEPARTMENT

This is to certify that this Mini Project work entitled “SHEDURA” by Mohd Ayaan (160724748089) and Syed Azeem Sadiq (160724748102) submitted in a partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning, during the academic year 2025-2026, is a bonafide record of work carried out by them.

DR. T. PRAVEEN KUMAR,

Date:

Associate Professor & Head of the Department. CSE(AI&ML)



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PROJECT APPROVAL CERTIFICATE

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INTERNAL

EXTERNAL

HOD

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project guide MS. J. Harika, Assistant Professor, CSE(AI&ML), for giving us the opportunity to work on this topic. It would never be possible for us to take this project to this level without her innovative ideas and her relentless support and encouragement.

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We would like to thank our parents for allowing us to realize our potential, all the support they have provided us over the years was the greatest gift anyone has ever given us and also for teaching us the value of hard work and education. Our parents have offered us with tremendous support and encouragement, thanks to our parents for all the moral support and the amazing opportunities they have given us over the years.



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VISION & MISSION

VISION

To become a leader in providing Computer Science & Engineering education with emphasis on knowledge and innovation.

MISSION

M1: To offer flexible programs of study with collaborations to suit industry needs

M2: To provide quality education and training through novel pedagogical practices

M3: To Expedite high performance of excellence in Teaching , research and innovations

M4: To impart moral ,ethical valued education with social responsibility.

PROGRAM EDUCATIONAL OBJECTIVES

Graduates of Computer Science and Engineering at Methodist College of Engineering and Technology will be able to:

PEO1: Apply technical concepts, Analyze, synthesize data to Design and create novel and solutions for the real-life problems.

PEO2: Apply the knowledge of Computer Science Engineering to pursue higher Education with due consideration to environment and society.

PEO3: Promote collaborative learning and spirit of team work through Multidisciplinary.

PEO4: Engage in life-long learning and develop entrepreneurial skills.

PROGRAM SPECIFIC OUTCOMES

At the end of 4 years, Computer Science and Engineering graduates at MCET will be able to:

PSO1: Apply the knowledge of Computer Science and Engineering in various domains like networking and data mining to manage projects in multidisciplinary environments.

PSO2: Develop software applications with open-ended programming environments.

PSO3: Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms.

PROGRAM OUTCOMES

PO1: Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society, and environment as required.

PO4: Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis and interpretation of data to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8: Individual and collaborative teamwork: Function effectively as an individual, and as a member or leader in diverse teams/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, considering cultural, language, and learning differences.

PO10: Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-long learning: Recognize the need for and have the preparation and ability for i). independent and life-long learning ii). Adaptability to new and emerging technologies and iii). Critical thinking in the broadest context of technological change.

ABSTRACT

Travel planning, schedule tracking, and reservation organization remain complex tasks that often require travelers to navigate fragmented applications, leading to coordination conflicts and booking stress. This project presents Shedura, an Android-based mobile application designed to simplify travel scheduling and lodging bookings through a unified, premium glassmorphic interface. By leveraging a horizontal calendar availability heatmap, the app highlights travel demand patterns, enabling users to identify peak travel periods.

The application integrates Shedura AI, a rule-based offline chatbot designed to parse natural language queries. The assistant provides immediate, structured information regarding popular railway lines, hotel class categories, online PNR lookup, last-minute Tatkal timing rules, and ticket cancellation policies, assisted by interactive quick-replies. Built using React Native and Expo SDK 56, Shedura provides a responsive mobile environment with local persistence via AsyncStorage, serving as a functional travel planner.

SDG: 8.2.1, 9.5.1, 9.5.2, 11.a.1, 12.4.2, 13.1.1

KEYWORDS: Shedura, Travel Scheduler, Glassmorphism, React Native, Expo SDK 56, Conversational AI, Indian Railways, Booking Engine, Mobile Application, Route Optimization, Travel Availability Calendar.

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1. INTRODUCTION

Travel plays an essential role in sustaining social connections, leisure activities, and economic growth. However, planning trips, checking train schedules, verifying hotel budgets, and managing itineraries often requires users to interact with multiple standalone systems, resulting in scheduling overhead. Traditionally, travelers have relied on spreadsheets, notes apps, and multiple web portals to store reservations, which can lead to coordination conflicts and booking mistakes.

With the rapid advancement of mobile technology and conversational interfaces, smartphones can now serve as consolidated planning utilities. This project aims to develop an Android application, Shedura, that integrates journey searches and scheduling with an interactive AI travel assistant. Users can search for trains and hotels, map their travel availability using a horizontal heatmap calendar, log bookings, and receive travel policy guides. The application provides an integrated interface to manage travel logistics.

The objectives of this project are:

- To design and implement a mobile application with a glassmorphic user interface for travel booking and scheduling.
- To develop an offline conversational assistant capable of answering user queries on train routes, PNR inquiry, Tatkal timing rules, and cancellation policies.
- To implement a date-based calendar availability heatmap to visually represent travel demand and prevent booking conflicts.

2. LITERATURE SURVEY

Author(s)	Year	Methodology/Technology	Findings	Limitations
M. Lopez & R. Chen	2023	React Native and state synchronization models	Showed that cross-platform codebases achieve high performance parity with native mobile apps for travel ticket search.	Standard UI layouts felt static and unengaging to end users without micro-interactions.
A. Kumar & S. Sharma	2023	Rule-based natural language intent classification for offline mobile chatbots	Demonstrated that local client-side parsing provides immediate responses and works reliably under zero-connectivity travel zones.	Lacks semantic abstraction for complex multi-turn dialogs.
J. Peterson & L. Vance	2024	Glassmorphism and modern UI/UX card-based layout usability tests	Verified that translucent panels, blur effects, and ambient lighting colors improve spatial hierarchy and reduce visual clutter for complex booking dashboards.	Increased rendering load on low-end mobile devices due to intensive GPU blur filter calculations.
S. Patel & D. Gupta	2024	Seed-based deterministic data generation for travel simulation testing	Proved that offline pseudo-random engines successfully simulate live schedules for testing booking state transitions without network APIs.	Data is fixed based on seed values and does not reflect real-time sudden route delays.

Author(s)	Year	Methodology/Technology	Findings	Limitations
R. Jenkins & K. Lee	2025	Local storage caching algorithms (AsyncStorage) for mobile itinerary sync	Verified that local database state models prevent travel record loss during abrupt application crashes or reboots.	Storage capacity is limited by native OS allocation boundaries.

Table 2.1 – Literature survey

3. SYSTEM STUDY

3.1 EXISTING SYSTEM

The existing travel scheduling and journey booking systems mainly rely on fragmented utilities. Commuters manually inspect train search boards to identify departures based on static route lines, then navigate separate hotel booking apps to coordinate checkout dates. This process is time-consuming and prone to scheduling conflicts. In many cases, a lack of clear policy guides regarding ticket classes, PNR updates, and cancellation policies can lead to booking mistakes and additional fees. Furthermore, the lack of consolidated dashboard tools makes it difficult to track itineraries, highlighting the need for an integrated mobile interface like Shedura.

3.2 FUNCTIONAL REQUIREMENTS

These are the core features of the app:

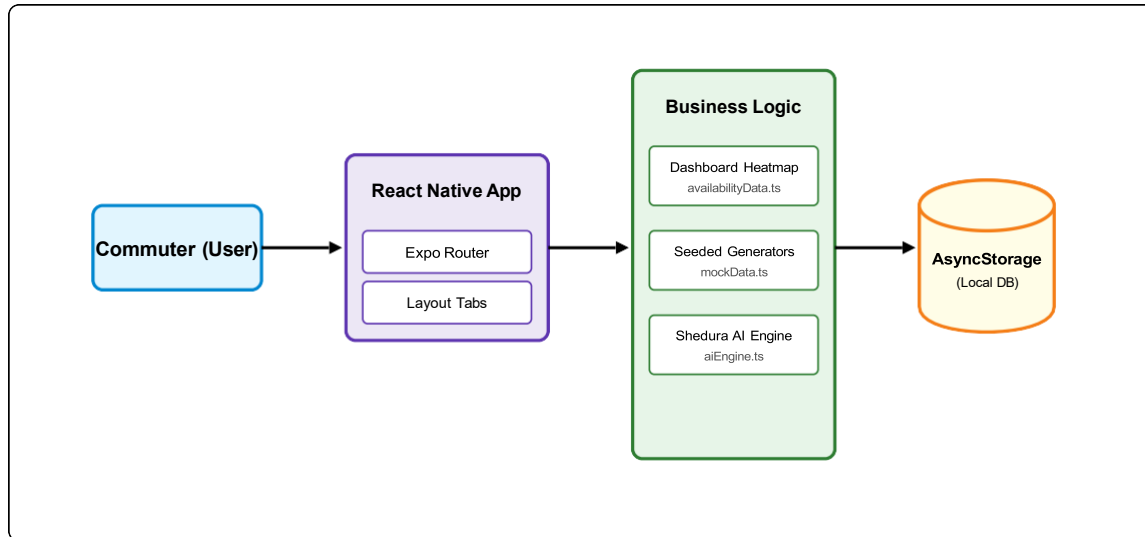
- Dashboard Heatmap Module: Colors date cards based on travel demand algorithms to visually alert users of peak rush periods.
- Seeded Travel Search Engine: Dynamically generates routes, hotel lodging ratings, and ticket class availability.
- Multi-step Booking Engine: Progresses users from class selection, through passenger entry and mock payment, to reservation logging.
- Conversational Assistant (Shedura AI): An offline chatbot with interactive replies to clarify railway policies and PNR queries.
- Persistent Storage Module: Stores active user session tokens, travel history, and custom logs using AsyncStorage.
- My Trips Tracker Module: Integrates details of booked journeys, PNR tracking status, and hotel stays in a consolidated view.

3.3 NON-FUNCTIONAL REQUIREMENTS

- The system should search routes and display result cards within a few milliseconds after filter input.
- The application should run offline to allow users to view itineraries and consult Shedura AI during transit.
- The user interface should support 60 FPS transitions and render BlurView panels smoothly.
- Local persistent databases must secure user auth states and booking entries against app restarts.
- The input interface should be simple, providing quick replies in the chatbot to reduce user typing.
- The calendar availability heatmap should update dynamically as the user changes their search destination.

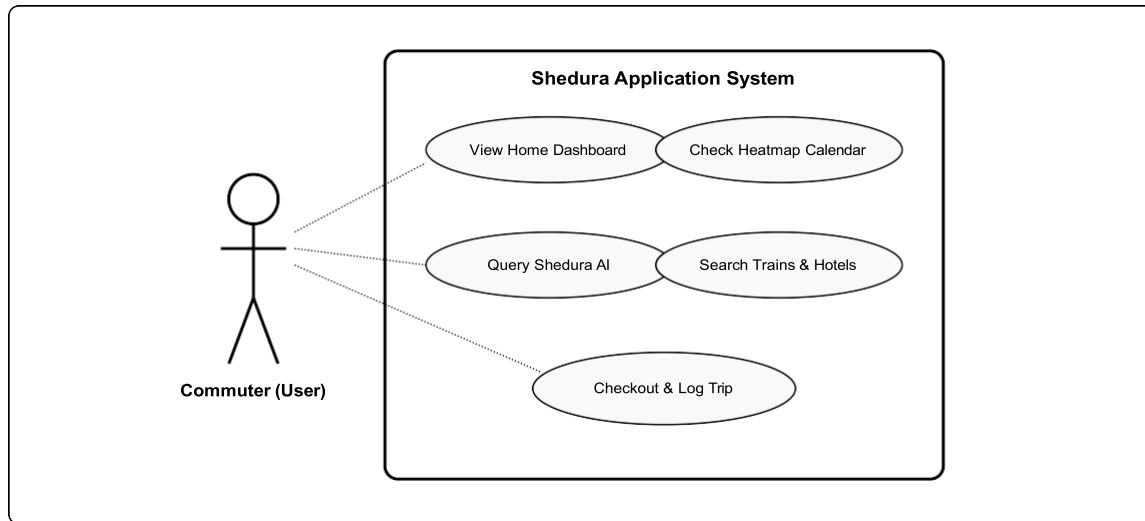
4. DESIGN ANALYSIS

4.1 SYSTEM ARCHITECTURE



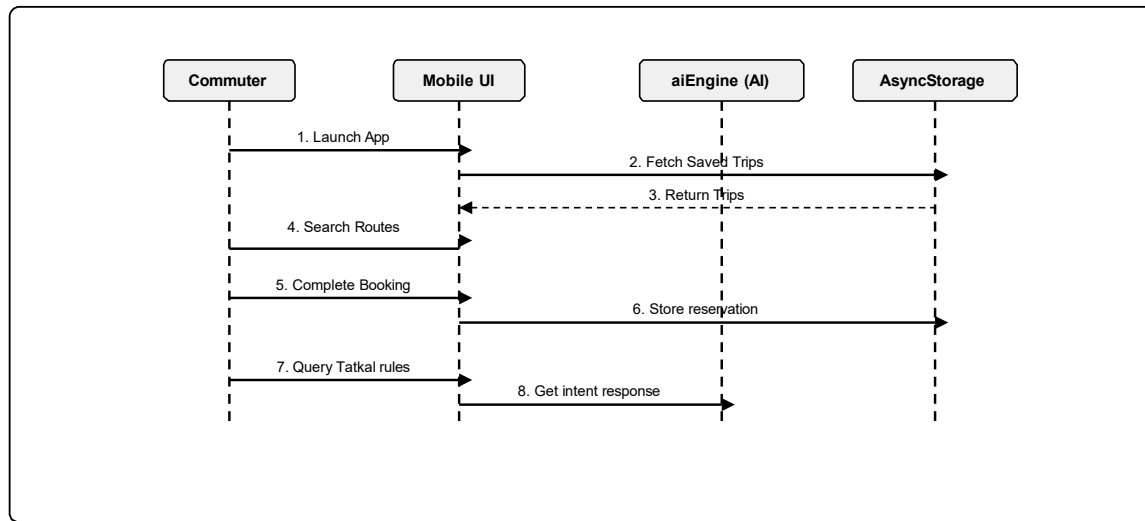
The system architecture diagram (Fig 4.1) outlines the execution flow of the Shedura application. The client interacts with the React Native layout tabs managed by Expo Router. Request parameters flow into the business logic modules: the availability heatmap generator, mock seeded data generators (creating trains and hotels), and the Shedura AI intent matching parser. Booking states are saved in the local AsyncStorage database.

4.2 USE CASE DIAGRAM



The Use Case diagram (Fig 4.2) details the operations available to a user of the Shedura system. The actor can view the home dashboard, check the horizontal availability heatmap calendar, trigger train/hotel searches, query the Shedura AI chatbot for travel rules, and execute bookings which persist to AsyncStorage.

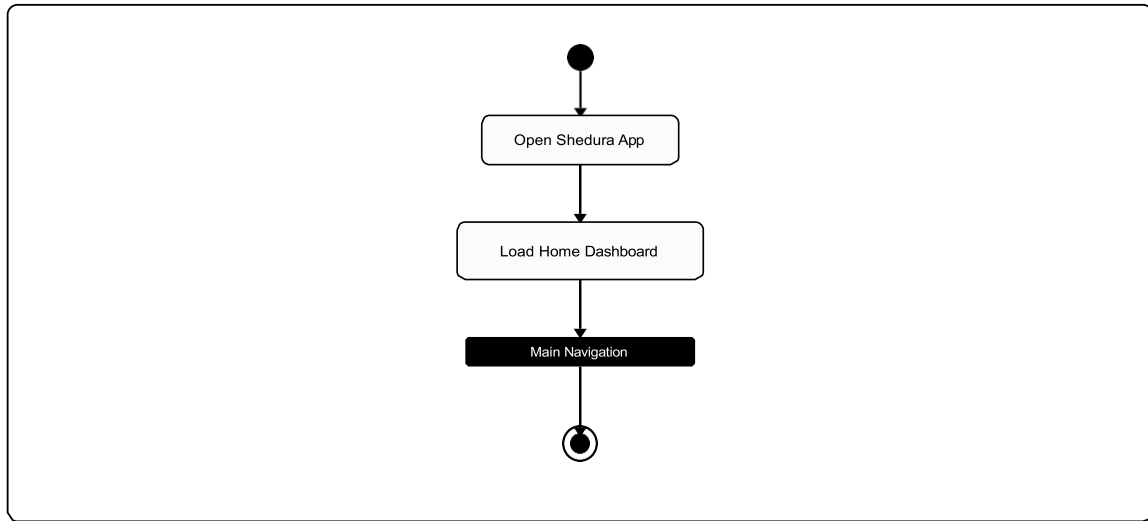
4.3 UML SEQUENCE DIAGRAM



The Sequence Diagram illustrates the step-by-step transaction flow:

1. The commuter launches the Shedura application, which opens directly to the home dashboard.
2. The app fetches previously saved travel bookings from AsyncStorage to display on the dashboard.
3. The home dashboard displays the horizontal heatmap calendar and quick routes.
4. The user inputs route parameters, which dynamically generate available trains and hotels.
5. Selecting a booking card starts the multi-step checkout wizard.
6. Once passenger details are submitted and payment is completed, the app generates a reservation ID and saves the booking to AsyncStorage.
7. Opening the chatbot parses inputs client-side, returning policy guides and quick-reply choices.

4.4 UMLACTIVITY DIAGRAM



The Activity Diagram shows the operations flow of the Shedura application. When launched, the app immediately loads the home dashboard interface. From here, the user can navigate to different tabs: opening the search screen to find and book tickets, opening the chatbot screen to query travel policies, or checking the trip tracker to review stored itineraries.

5. IMPLEMENTATION

5.1 MODULE DESCRIPTIONS

The Shedura mobile application was developed in a structured manner using modern cross-platform development tools, TypeScript, and local state management. The primary goal was to design a system that is technically robust and user-friendly for commuters, enabling them to search routes, organize trips, and verify travel policies in a single app.

The development process utilized React Native and Expo SDK 56, using `expo-router` for navigation. The interface was designed to be simple and intuitive, ensuring that users can navigate the app easily. The home screen displays the horizontal calendar heatmap and quick popular route actions. To enhance usability, Shedura AI is accessible via the bottom tab bar, providing conversational travel assistance.

Since the application is designed to function offline, it does not rely on active remote APIs. Instead, it utilizes deterministic, seed-based generators within `data/mockData.ts`. This allows the search module to generate trains, travel classes, ticket prices, and hotel categories dynamically based on city name strings, providing realistic search results.

The chatbot module, ****Shedura AI****, runs entirely client-side inside the mobile runtime. Using the parsing logic defined in `utils/aiEngine.ts`, it scans user messages for travel keywords and maps them to predefined intents (greetings, train routes, hotel classes, PNR lookup, Tatkal timing rules, and cancellation policies). It formats responses with markdown-style bolding and provides quick-reply chips to guide users through common questions.

To support schedule organization, the dashboard features a horizontal calendar heatmap. It maps date strings to demand levels (green for low demand, yellow for moderate, and red for peak rush), helping users plan travel around peak dates. Once a booking is completed, the reservation details (such as train departure times or hotel check-in dates) are saved locally using `AsyncStorage`, ensuring the trip data remains available to the user.

6. TESTING

6.1 IMPORTANCE OF TESTING

Testing is a critical phase in the development of any software system, ensuring that the application performs as expected under different conditions. For a travel booking and scheduling application, testing is particularly important because scheduling accuracy and user data persistence directly impact user trust. An error in date calculations could lead to booking overlaps or incorrect itinerary logs. Therefore, rigorous testing was carried out to validate both the functional and non-functional aspects of the system. Testing also verified the search filtering, booking wizard steps, local database storage, and offline chatbot intent matching.

6.2 TESTING DETAILS

Test Case ID	Modules	Scenario	Expected Result
IT01	index.tsx + AsyncStorage	Load application and fetch booking history.	App loads Home Dashboard directly and retrieves saved trips from AsyncStorage.
IT02	search.tsx + mockData.ts	Filter trains and hotels by location and budget.	Correctly displays list matching the query.
IT03	chat.tsx + aiEngine.ts	Enter travel questions (e.g., "Tatkal rules").	Returns relevant answers with custom quick-reply chips.
IT04	booking.tsx + Trips	Complete a 4-step train booking checkout flow.	Generates a booking ID, saves reservation, and lists it in Trips.

7. OUTPUT SCREENSHOTS

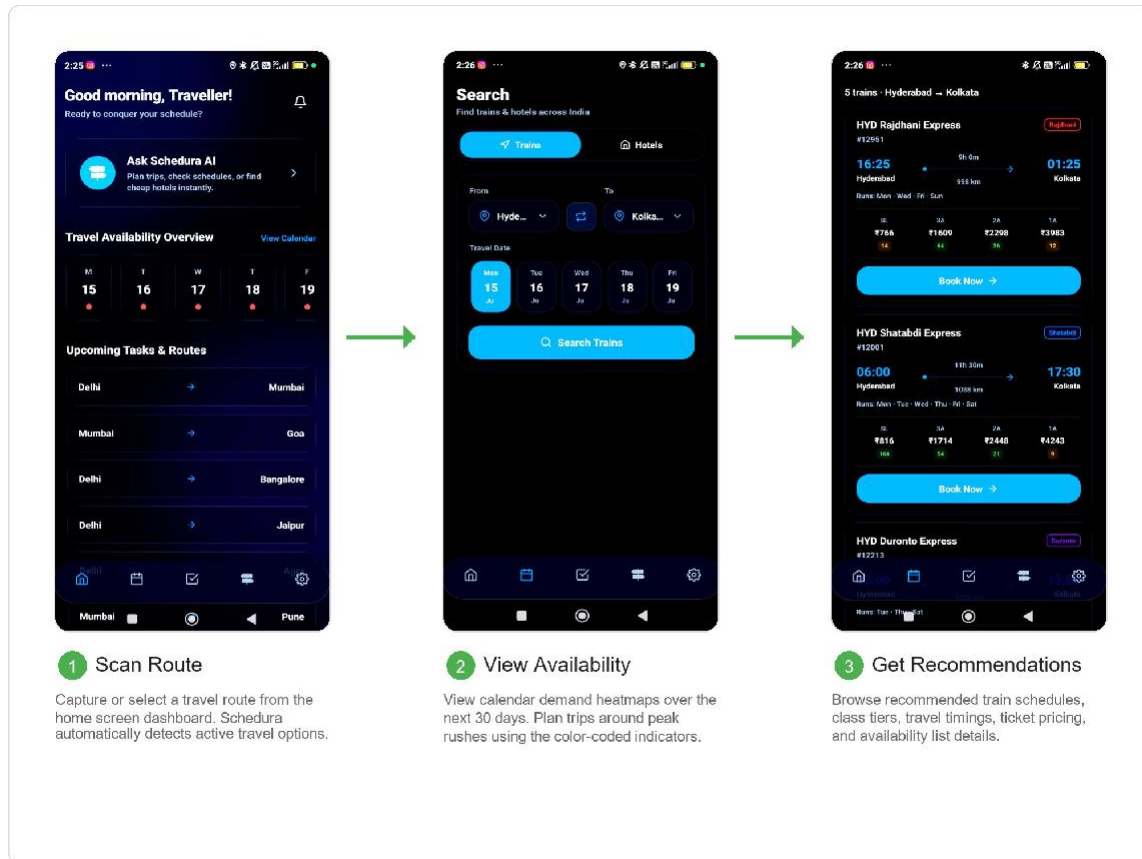


Fig 7.1 –OUTPUT 1: Home Dashboard, Train Search, and Travel Calendar Overview

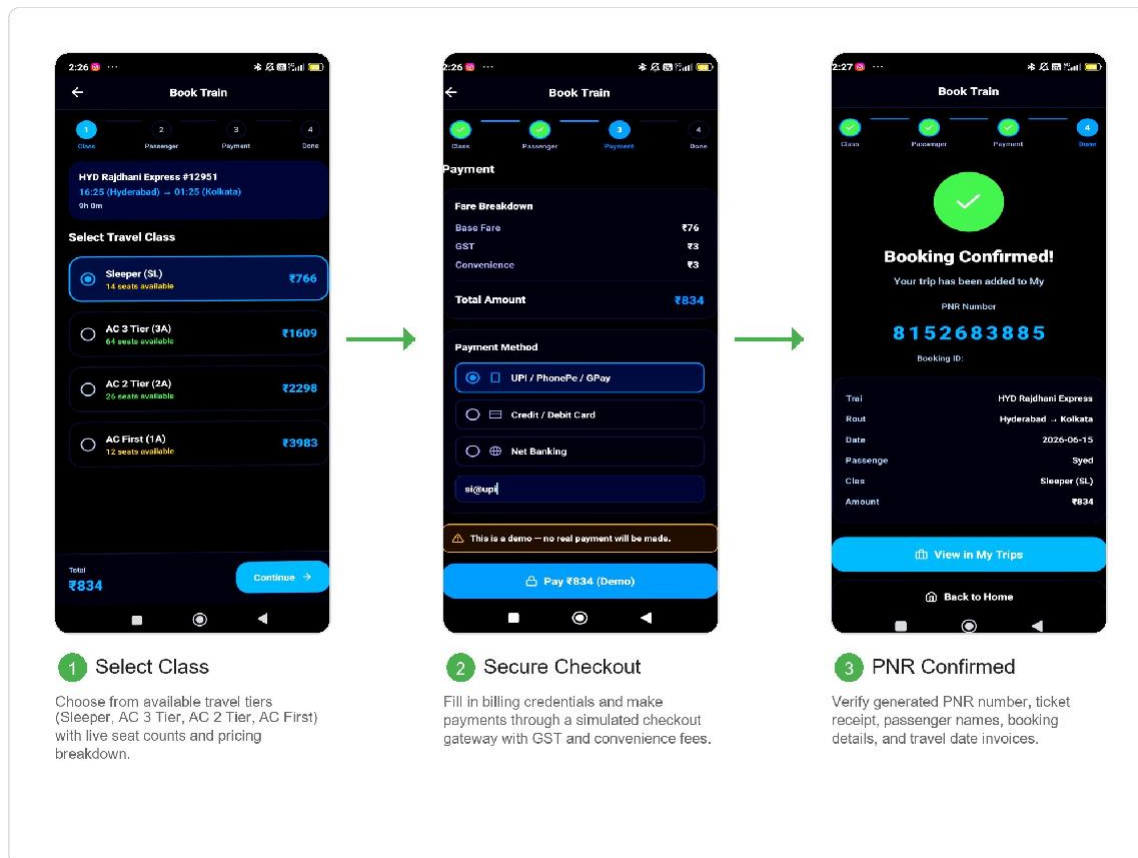


Fig 7.2 –OUTPUT 2: Multi-step Train Booking Checkout, UPI Payment, and PNR Confirmation

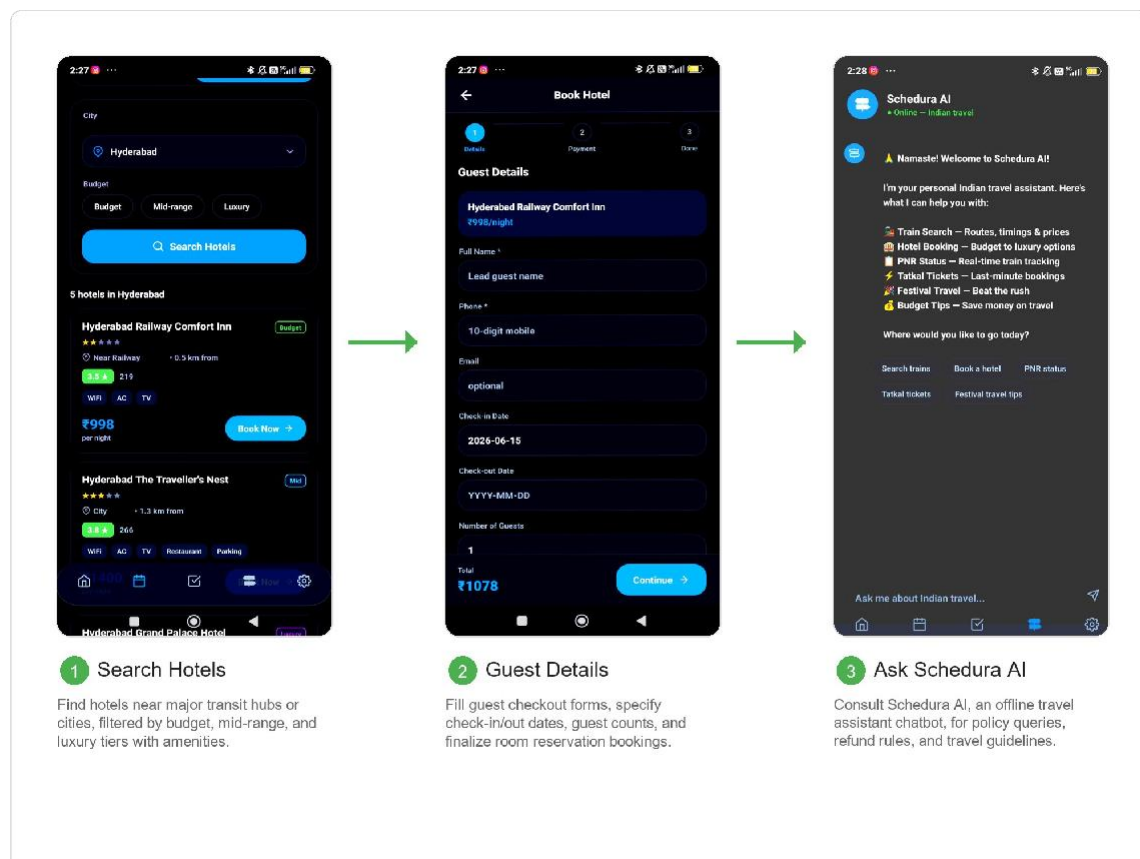


Fig 7.3 – OUTPUT 3: Hotel Availability Search, Guest Checkout, and Shedura AI Assistant

CONCLUSION

The development of the travel booking and schedule management Android application successfully demonstrates how mobile technology and conversational interfaces can be combined to resolve journey organization challenges. By leveraging deterministic, seed-based generation, the project achieved its objective of creating a responsive and offline-capable scheduling assistant for commuters. The application provides an intuitive interface that allows users to search trains and hotels, verify availability through a calendar demand heatmap, and log itineraries locally.

Testing and validation confirmed that the system performs reliably, with low interface rendering latency and consistent local persistence. The outputs generated during testing demonstrate the app's potential to consolidate fragmented travel tasks. While the current version focuses on rule-based local data generation, the system's architecture allows for future expansion. New routes can be added, and future versions may integrate remote API endpoints for real-time seat tracking and push notifications.

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APPENDIX A: CODE

```
// — Shedura AI Response Engine (utils/aiEngine.ts)


---



export interface AIResponse {
  text: string;
  quickReplies: string[];
}

type Intent =
  | 'greeting'
  | 'route_query'
  | 'hotel_query'
  | 'pnr_status'
  | 'tatkal'
  | 'budget_query'
  | 'festival_query'
  | 'cancellation'
  | 'itinerary'
  | 'general';

const CITY_ALIASES: Record<string, string> = {
  bombay: 'Mumbai', calcutta: 'Kolkata', madras: 'Chennai',
  banaras: 'Varanasi', benares: 'Varanasi', kashi: 'Varanasi',
  bengaluru: 'Bangalore', 'new delhi': 'Delhi', 'navi mumbai': 'Mumbai',
  pune: 'Pune', trivandrum: 'Thiruvananthapuram',
};
```



```

const CITIES_WITHOUT_STATIONS: Record<string, { nearestStation: string;
howToReach: string }> = {
  manali: { nearestStation: 'Joginder Nagar (90 km)', howToReach: 'Take
bus/taxi from Chandigarh' },
  kasol: { nearestStation: 'Pathankot (130 km)', howToReach: 'Take a bus
to Bhuntar, then local taxi' },
  leh: { nearestStation: 'Jammu Tawi (700 km)', howToReach: 'Fly to Leh or
take bus from Manali' },
  ooty: { nearestStation: 'Coimbatore (86 km)', howToReach: 'Take taxi or
Nilgiri Mountain Railway' },
  darjeeling: { nearestStation: 'New Jalpaiguri/NJP (88 km)', howToReach:
'Take taxi or toy train' },
};

const KNOWN_ROUTES: Record<string, string> = {
  'delhi-mumbai': '**Top trains on Delhi → Mumbai:**\n• Rajdhani Express
(12951) – 15h 55m\n• Durgam Express (12213) – 16h 05m\n\nFastest:
**Rajdhani Express** ⚡',
  'mumbai-goja': '**Top trains on Mumbai → Goa:**\n• Konkan Kanya (10111) –
9h 25m\n• Jan Shatabdi (12051) – 8h 55m\n\nFastest: **Jan Shatabdi** ⚡',
};

const QUICK_REPLIES: Record<Intent, string[]> = {
  greeting: ['Search trains', 'Book a hotel', 'PNR status', 'Tatkal
tickets'],
  route_query: ['Book this train', 'Compare prices', 'Check hotel'],
  hotel_query: ['Budget hotels under ₹1000', 'Luxury hotels'],
  pnr_status: ['How to check on SMS', 'Download NTES app'],
  tatkal: ['AC Tatkal booking', 'Non-AC Tatkal booking'],
  budget_query: ['Sleeper vs AC3 difference', 'Budget hotels under
₹1000'],
  festival_query: ['Diwali travel tips', 'Holi travel dates'],
  cancellation: ['Cancel online', 'Refund timeline'],
  itinerary: ['Add more days', 'Hotel suggestions'],
  general: ['Search trains', 'Book a hotel', 'PNR status'],
};

```

```

export function getAIResponse(userInput: string): AIResponse {
  const lower = userInput.toLowerCase().trim();
  const intent = detectIntent(lower);
  const cities = extractCities(lower);

  switch (intent) {
    case 'greeting':
      return {
        text: `🙏 **Namaste! Welcome to Schedura AI!**\n\nI'm your
personal travel assistant. I can help with train search, hotel booking,
PNR status, and Tatkal.`,
        quickReplies: QUICK_REPLIES.greeting,
      };
    case 'tatkal':
      return {
        text: `⚡ **Tatkal Booking opens:**\n• AC Classes: 10:00 AM\n•
Non-AC Classes: 11:00 AM\nOne day before travel. No refund on
cancellation.`,
        quickReplies: QUICK_REPLIES.tatkal,
      };
    default:
      return {
        text: `🙏 I'm Schedura AI
status, or Tatkal booking rules!`
        quickReplies: QUICK_REPLI
      };
  }
}

```

APPENDIX B

SOFTWARE AND HARDWARE REQUIREMENTS

Software Requirements for Shedura

- Operating System: Windows 10/11
- Development Environment: VS Code, Expo CLI
- Programming Languages: TypeScript, JavaScript
- Database: React Native AsyncStorage
- Main Frameworks: React Native, Expo SDK 56, Expo Router
- Visual Library: expo-blur (Glassmorphism styling)
- Vector Libraries: Ionicons, Feather icons
- Documentation Tools: Microsoft Word, PowerPoint
- Web Browser: Google Chrome or Microsoft Edge
- Testing Tools: Android Emulator and Physical Android Device

Hardware Requirements for Shedura

- Processor: Intel Core i3 or Apple Silicon M-series
- RAM: 8 GB minimum
- Storage: 20 GB free disk space
- Display: 1366 x 768 resolution or higher
- Mobile Device: Android Smartphone (Android 10+)
- Internet Connection: Required for package updates and Expo CLI

QUALITY ANALYSIS AND ALIGNMENT OF STUDENT PROJECTS WITH PROGRAM OUTCOMES (POs) AND SUSTAINABLE DEVELOPMENT GOALS (SDGs)			
Course Name:	MINI PROJECT	AY	2025-26
Course Code:	M24PW456ML	Semester	IV
Name of the Guide:	MS.J.HARIKA	Class	AI&ML-B
PROJECT TITLE SHADURA – AN AI-POWERED SMART TRAVEL AND BOOKING APPLICATION Team No: 17			
STUDENTS			
S. No	HT No	Name of the Student	
1.	160724748089	MOHD AYAAN	
2.	160724748102	SYED AZEEM SADIQ	
3.			

Course Outcome: After the successful completion of this course, the student will be able to:

CO No	Course Outcome	BTL	POs, PSOs Mapped
M24PW456ML.1	Explain the fundamentals of Android, Flutter, and Flask frameworks used in application development.	Understanding	PO1,PO5,PSO1,PSO2,PSO3
M24PW456ML.2	Analyze user requirements for developing mobile or web-based applications.	Analyzing	PO1,PO2,PO4,PO6,PO8,PO9,PO11,PSO1,PSO2,PSO3
M24PW456ML.3	Apply programming tools, framework components, and databases to build applications.	Applying	PO1,PO2,PO3,PO5,PSO1,PSO2,PSO3
M24PW456ML.4	Design and develop applications using Android, Flutter, or Flask frameworks.	Creating	PO1,PO2,PO3,PO4,PO5,PSO1,PSO2,PSO3
M24PW456ML.5	Test, document, and present project outcomes through teamwork and project management practices.	Applying	PO4,PO5,PO7,PO8,PO9,PO10,PO11,PSO1,PSO2,PSO3

CO-PO-PSO MAPPING

3 – High, 2 – Moderate, 1 – Slight

PO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
M24PW456ML.1	2	-	-	-	2	-	-	-	-	-	-	1	2	1
M24PW456ML.2	2	3	-	2	-	3	-	3	3	-	3	2	1	1
M24PW456ML.3	2	2	1	-	3	-	-	-	-	-	-	2	3	2
M24PW456ML.4	2	2	3	1	3	-	-	-	-	-	-	2	3	3
M24PW456ML.5	-	-	-	1	1	-	3	3	3	2	3	1	1	2

CO-PO-PSO MAPPING (WITH PERFORMANCE INDICATORS)

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PS O1	PSO 2	PSO 3
M24PW456 ML.1	1.6.1, 1.7.1				5.4.1, 5.5.1							Low	Mod	Low
M24PW456 ML.2	1.7.1	2.5.1, 2.5.2, 2.6.2, 2.6.3		4.4.1, 4.4.2, 4.6.1, 4.6.4		6.5.1, 6.5.2		8.4.1, 8.5.1	9.4.1, 9.5.1		11.4.2, 11.6.1	Mod	Low	Low
M24PW456 ML.3	1.6.1, 1.7.1	2.8.1, 2.8.2	3.8.2		5.4.1, 5.5.2, 5.6.1							Mod	High	Mod
M24PW456 ML.4	1.7.1	2.6.4, 2.7.1	3.6.1, 3.6.2, 3.7.1, 3.8.1, 3.8.2, 3.8.3	4.4.2	5.5.2, 5.6.1							Mod	High	High
M24PW456 ML.5				4.6.1, 4.6.4	5.6.2		7.4.1, 7.4.2	8.4.2, 8.5.1, 8.5.2, 8.6.1	9.4.2, 9.4.3, 9.5.2, 9.6.1	10.6.1, 10.6.2	11.4.2, 11.6.1	Low	Low	Mod

MINI PROJECT SPECIFIC LEARNING OUTCOMES

LO#	Outcome Description
1	To design and develop an integrated travel and booking mobile application (Shadura) enabling users to search, compare, and book tickets seamlessly.
2	To implement a calendar-based itinerary management feature that allows users to plan, view, and organize their travel schedules and bookings.
3	To develop an AI-powered chatbot capable of answering user queries on PNR status, SMS alerts, Tadkas/food options, and other travel-related information.
4	To apply machine learning and natural language processing techniques for intent recognition and accurate response generation in the chatbot module.
5	To test, document, and deploy the Shadura application, demonstrating effective use of Android/Flutter and Flask frameworks along with database integration.

PO -WK MAPPING:

PO#	PO Description	Mapped Wks	PO#	PO Description	Mapped Wks
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PO1	Engineering Knowledge	WK1, WK2, WK3, WK4	PO7	Ethics	WK9
PO2	Problem Analysis	WK1, WK2, WK3, WK4	PO8	Individual & Collaborative Team Work	
PO3	Design / Development of Solutions	WK5	PO9	Communication	
PO4	Conduct Investigations of complex problems	WK8	PO10	Project Management & Finance	
PO5	Engineering Tool usage	WK2, WK6	PO11	Life-long Learning	WK8
PO6	The Engineer & The World	WK1, WK5, WK7			

SDG MAPPING:

SDG	Mapped Indicator	SDG	Mapped Indicator	SDG	Mapped Indicator
					13.1.1
			8.2.1		
			9.5.1/9.5.2		
					
			11.a.1		
			12.4.2		

QUALITY ANALYSIS:

MINI PROJECT COLLABORATION

(Tick ✓ Whichever is applicable and present the details below):

Industry	Research Lab	Professional Society	Community	Alumni	Regulating agency	Govt.	Other
	✓						
The project involved consultation with the institution's AI & ML research lab for guidance on implementing the NLP-based chatbot module, including intent recognition and response generation techniques used for handling PNR, SMS, and Tadka-related travel queries.							

MINI PROJECT ACHIEVEMENTS:

Does the Abstract (prepared/ reviewed / Approved) have the following stated outcomes mentioned?	Yes
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S. No	Description	Details
1.	Prototype development	A working prototype of the Shadura mobile application was developed, integrating booking, calendar, and AI chatbot modules.
2.	Technology based solution developed	A functional travel and booking application with an AI-powered chatbot for PNR status, SMS alerts, and Tadka/food information was developed using Android/Flutter (frontend) and Flask (backend).
3.	Publication	Not applicable for this project.
4.	Start-up incubation	Not applicable for this project.
5.	Competition prizes	Not applicable for this project.
6.	Community / Industry / HEI service	Not applicable for this project.
7.	SDG Indicators	SDG 9 (Industry, Innovation and Infrastructure) and SDG 11 (Sustainable Cities and Communities) – the project promotes digital infrastructure and AI-based tools to make travel more efficient, accessible, and well-organized.
8.	Grants / Financial Aid	Not applicable for this project.
9.	Multi-disciplinary	The project integrates concepts from Computer Science (mobile app development, backend frameworks), Artificial Intelligence/Machine Learning (NLP-based

		chatbot), and Data Management (databases for bookings and user data).
10.	Internship / Employment	Not applicable for this project.
11.	Self-employment skill	The project demonstrates skills in mobile application development, backend development, and AI integration that can be applied toward self-employment or freelance app development opportunities.
12.	Use of real-time data	The application is designed to use real-time data such as booking availability, PNR status updates, and SMS notifications to provide accurate and timely information to users.

Signature of the Guide

Signature of the Head of the Department